

Kalman Filter

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Introduction

Parametrization

There are different methods to parametrize models:

- Calibration
- Matching Moments
- Estimation
 - Maximum Likelihood
 - Bayesian Estimation

Calibration

In a nutshell:

- Giving values to structural parameters in such way to replicate real world observations,
- Use different values to perform sensitivity analysis.

The problem we calibration:

- Difficult to chose the right values for the parameters,
- I could lead (sometimes) to arbitrary values with very little economic sens.

Matching Moments

Similarly to calibration:

- Using values for parameters by choosing a set of moments that we would like our model to match,
- Performing a sensitivity analysis to judge the model fit by investigating the set of moments.

Contrary to calibration:

- estimated (different than the estimation we will use for Kalman filter)
- Allow for hypothesis testing

Estimation

Full information method as we have to specify an entire distribution:

- **Maximum Likelihood:** The model fit is then based on implied likelihood function (i.e. how consistent with the distribution assumption we have made).
- **Bayesian estimation:** Need to specify a prior (will be discussed further on Bayesian estimation techniques).

Maximum Likelihood

Kalman Filter

The Kalman filter was initially developed for engineering control problems (e.g. NASA spatial program)

- Following Hamilton notation we will set up the Kalman filter based on general time series model,
- Then the time series model will be linked to DSGE models.

Kalman Filter: Times Series Model

The model:

$$y_t = M'_t x_t + w_t, \quad E(w_t, w'_t) = R, \quad (1)$$

$$x_{t+1} = F_t x_t + v_{t+1}, \quad E(v_t, v'_t) = Q. \quad (2)$$

- y_t is an observed variable, while x_t is not observed,

⇒ Kalman filter allows us to retrieve an estimate of x_t .

Kalman Filter: Some initial hypotheses

- we assume that w and v are uncorrelated (we can easily extend the model to account for cross correlations),
- we assume for now that H , F , R , and Q are known and time unvarying (without a loss of generality). (Later on we will show how to estimate them),
- we need to specify an initial condition for x_t as it is a forward looking variable: x_1 with mean $\hat{x}_1$ and variance $P_{1|0}$,
- observation (w_t) and state (v_t) disturbances are:
 - uncorrelated over time,
 - uncorrelated with each other (at all leads and lags)
 - orthogonal to x_1

Deriving the Kalman Filter

The main idea of Kalman filter is to give a prediction about a variable of interest (e.g. potential GDP) giving that we observed a series of a variable values in the past (e.g. GDP):

$$\hat{x}_{t+1} = \hat{E}(x_{t+1}|Y_t), \text{ where: } Y_t = (y'_t, y'_{t-1}, \dots, y'_1). \quad (3)$$

Now the question is how well our prediction?

- Thus, a need to specify a loss function to be able to evaluate the forecast:

$$\min E(x_{t+1} - \hat{x}_{t+1|t})^2 \quad (4)$$

- $\hat{x}_{t+1|t}$ is then the projection of x_{t+1} on its regressors.

Deriving the Kalman Filter

- Let's recall that any projection of x on Y is a function of $\hat{E}[x|Y] = f(x)$.
- Assuming then both a linear form for the projection where $f(x) = \alpha + \beta x$ and the existence of both first and second moments¹

We then solve the minimization problem (in the general framework before going back to our time series model) as follow:

$$\beta = \frac{\text{Cov}(x, Y)}{\text{Cov}(Y, Y)}, \quad (5)$$

$$\alpha = \bar{x} - \beta \bar{Y}, \quad (6)$$

$$\hat{x} = \hat{E}(x|Y) = \bar{x} + \frac{\text{Cov}(x, Y)}{\text{Cov}(Y, Y)}(Y - \bar{Y}). \quad (7)$$

¹ $\bar{x} = E(x)$, $\text{Cov}(x, x) = E[(x - \bar{x})(x - \bar{x})']$, $\bar{Y} = E(Y)$, $\text{Cov}(Y, Y) = E[(Y - \bar{Y})(Y - \bar{Y})']$, and $\text{Cov}(x, Y) = E[(x - \bar{x})(Y - \bar{Y})']$.

Deriving the Kalman Filter

We should not confuse projection with regression (e.g. OLS). Although the expression we are seeking to minimize are closely similar, OLS seeks to investigate a causality, while the projection is solely focused on forecasting.

Deriving the Kalman Filter

Jumping back to our model!

The main idea is to write down the model recursively:

- As we have a value for $x_{1|0}$, and the observation for Y_t (i.e. $y_1, \dots, y_T$), we can then have an update for $\hat{x}_{1|1}$ and then have a forecast value for $\hat{x}_{2|1}$. Then we keep repeating this procedure.

Thus,

$$\begin{aligned}\hat{x}_{t|t} &= \hat{E}(x_t | Y_t) \\ &= \hat{x}_{t|t-1} + E[(x_t - \hat{x}_{t|t-1})(y_t - \hat{y}_{t|t-1})'] * \\ &\quad E[(y_t - \hat{y}_{t|t-1})(y_t - \hat{y}_{t|t-1})']^{-1}(y_t - \hat{y}_{t|t-1}).\end{aligned}\quad (8)$$

Deriving the Kalman Filter

Using the model equations, we can easily rewrite and simplify the co-variance, the variance, and error terms, respectively, as following:

$$\begin{aligned}
 E[(x_t - \hat{x}_{t|t-1})(y_t - \hat{y}_{t|t-1})'] &= E[(x_t - \hat{x}_{t|t-1})(M'(x_t - \hat{x}_{t|t-1}) + w_t)'] \\
 &= E[(x_t - \hat{x}_{t|t-1})((x_t - \hat{x}_{t|t-1})'M)] \\
 &= P_{t|t-1}M \quad (9)
 \end{aligned}$$

$$\begin{aligned}
 E[(y_t - \hat{y}_{t|t-1})(y_t - \hat{y}_{t|t-1})'] &= E[M'(x_t - \hat{x}_{t|t-1})(x_t - \hat{x}_{t|t-1})M] \\
 &\quad + E[w_t w_t'] \\
 &= M'P_{t|t-1}M + R \quad (10)
 \end{aligned}$$

$$y_t - \hat{y}_{t|t-1} = y_t - M'\hat{x}_{t|t-1} \quad (11)$$

writing down the final form of the Kalman Filter

The state variable:

The update step:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + P_{t|t-1} M (M' P_{t|t-1} M + R)^{-1} (y_t - M' \hat{x}_{t|t-1}) \quad (12)$$

The forecast step:

$$\begin{aligned} \hat{x}_{t+1|t} &= \hat{E}(x_{t+1} | Y_t) \\ &= F \hat{E}(x_t | Y_t) \\ &= F \hat{x}_{t|t-1} + F P_{t|t-1} M (M' P_{t|t-1} M + R)^{-1} (y_t - M' \hat{x}_{t|t-1}) \end{aligned} \quad (13)$$

writing down the final form of the Kalman Filter

Specifying the variance matrix recursion formula:

The update step:

$$\begin{aligned}
 P_{t|t} &= E[(x_t - \hat{x}_{t|t})(x_t - \hat{x}_{t|t})'] \\
 &= E[(x_t - \hat{x}_{t|t-1})(x_t - \hat{x}_{t|t-1})'] - E[(x_t - \hat{x}_{t|t-1})(y_t - \hat{y}_{t|t-1})'] * \\
 &\quad E[(y_t - \hat{y}_{t|t-1})(y_t - \hat{y}_{t|t-1})']^{-1} * \\
 &\quad E[(y_t - \hat{y}_{t|t-1})(x_t - \hat{x}_{t|t-1})'] \\
 &= P_{t|t-1} - P_{t|t-1} M (M' P_{t|t-1} M + R)^{-1} M' P_{t|t-1} \quad (14)
 \end{aligned}$$

The forecast step:

$$\begin{aligned}
 P_{t+1|t} &= E[(x_{t+1} - \hat{x}_{t+1|t})(x_{t+1} - \hat{x}_{t+1|t})'] \\
 &= E[(F x_t + v_{t+1} - F \hat{x}_{t|t})(F x_t + v_{t+1} - F \hat{x}_{t|t})'] \\
 &= F E[(x_t - \hat{x}_{t|t})(x_t - \hat{x}_{t|t})'] F' + E[v_{t+1} v_{t+1}'] \\
 &= F P_{t|t} F' + Q \quad (15)
 \end{aligned}$$

Summary of the Kalman Filter

The update step:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + P_{t|t-1}M(M'P_{t|t-1}M + R)^{-1}(y_t - M'\hat{x}_{t|t-1}) \quad (16)$$

$$P_{t|t} = P_{t|t-1} - P_{t|t-1}M(M'P_{t|t-1}M + R)^{-1}M'P_{t|t-1} \quad (17)$$

The forecast step:

$$\hat{x}_{t+1|t} = F\hat{x}_{t|t-1} + FP_{t|t-1}M(M'P_{t|t-1}M + R)^{-1}(y_t - M'\hat{x}_{t|t-1}) \quad (18)$$

$$P_{t+1|t} = FP_{t|t}F' + Q \quad (19)$$

Maximum Likelihood (ML)

Estimation

- We can also estimate all the parameter we assumed previously fixed (i.e. $\varkappa = \{M, F, Q, R\}$) using ML,
- Assuming $x_{1|0}$ and both error terms are Gaussian, then the distribution of y_t / Y_t is also Gaussian
- We then given values of $\varkappa$ we can derive the mean and variance of y ,
- And as we know the distribution of y , this allows for computing the likelihood of the observations y_t
- Finally, we just maximize the log-likelihood with respect to the parameters $\varkappa$

Liking the Time Series Model to DSGE

DSGE Example: Neo-classical Growth Model

Household: The household maximize their lifetime utility. They consume and invest in capital.

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}} E_0 \sum_{t=0}^{\infty} \beta^t (\log(c_t)), \\ c_t + k_{t+1} = y_t + (1 - \delta)k_t \end{aligned} \quad (20)$$

(with initial value for k_0 known.)

Production: The production follow a classical cobb-douglas function and is subject to exogenous technology shocks

$$y_t = a_t k_t^\alpha \quad (21)$$

$$a_t = 1 - \rho + \rho a_{t-1} + \epsilon_t \quad (22)$$

with $\epsilon_t \sim N(0, \sigma^2)$ and a_0 known.

DSGE Example: Neo-classical Growth Model

Solving the modeling (i.e. finding the policy functions)

The first order conditions read:

$$c_t^{-1} = E_t[\beta c_{t+1}^{-1}(\alpha z_{t+1} k_t^\alpha (1 - \delta) + 1 - \delta)] \quad (23)$$

$$c_t + k_t = a_t k_{t-1}^\alpha + (1 - \delta) k_{t-1} \quad (24)$$

Thus, the policy function can be written as following:

$$k_t = k_{ss} + \alpha_{kk}(k_{t-1} - k_{ss}) + \alpha_{ka}(a_t - a_{ss}) \quad (25)$$

$$a_t = (1 - \rho)a_{ss} + \rho a_{t-1} + \epsilon_t \quad (26)$$

where $a_{ss} = 1$.

- $\alpha_{kk}, \alpha_{ka}, k_{ss}$ are functions which depend on the structural parameters $\varkappa = \{\alpha, \beta, \delta, \rho, \sigma, a_0\}$.

DSGE Example: Neo-classical Growth Model

Estimating the model: Using the state-space form

- Using ML we can estimate the model
- To do so we need to specify the likelihood function, thus the Kalman filter M and F matrices equivalents.

The state-space form (let's recall the general case)

$$\begin{aligned}y_t &= M'_t x_t + w_t, \quad E(w_t, w'_t) = R, \\x_{t+1} &= F_t x_t + v_{t+1}, \quad E(v_t, v'_t) = Q,\end{aligned}$$

DSGE Example: Neo-classical Growth Model

Estimating the model: Using the state-space form

Our Neo-classical Growth model:

$$\begin{bmatrix} k_t - k_{ss} \\ a_{t+1} - a_{ss} \end{bmatrix} = \begin{bmatrix} \alpha_{kk} & \alpha_{ka} \\ 0 & \rho \end{bmatrix} \begin{bmatrix} k_{t-1} - k_{ss} \\ a_t - a_{ss} \end{bmatrix} + \begin{bmatrix} 0 \\ \epsilon_{t+1} \end{bmatrix}$$

$$k_t - k_{ss} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} k_{t-1} - k_{ss} \\ a_t - a_{ss} \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix}$$

The work is done, now we just use the previous section results to compute the Kalman filter, while using ML through maximizing the log-likelihood function with respect to the parameters $\mathcal{K}$.

DSGE Example: Neo-classical Growth Model

Estimating the model: Using the state-space form

The log-likelihood function for the Gaussian distribution assumed in the beginning reads as following:

$$\mathcal{L}(Y_t|\mathcal{K}) = \log(f(y_0)) + \sum_{t=1}^T \log(f(y_t|Y_{t-1}; \mathcal{K})) \quad (27)$$

which in this example would be:

$$\log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{\epsilon(\mathcal{K})^2}{2\sigma^2} \quad (28)$$

Exercise

Constructing Kalman Filter

We will be using Kalman filter to estimate the aggregate matching function using ML.

Using the aggregate matching function, which allows for capturing the relationship between the number of new jobs (hires) H_t , and the number of unemployed U_t as well as the number of vacancies present in the economy V_t , is a wide spread technique to measure unemployment. We characteristic the functional form following a classical Cobb Douglas constant returns to scale function.

$$H_t = MU_t^{1-\rho} V_t^\rho \quad (29)$$

- where, M the constant level parameter referring to the degree of mismatch or matching efficiency,
- ρ the elasticity of matches with respect to the number of unemployed,

Constructing Kalman Filter

The goal of this exercise is to attempt an estimation of the aggregate matching function, while allowing match efficiency to fluctuate over time, in order to investigate to role of the mismatch in the unemployment spike during the Great Recession.

First Step:

Estimating the aggregate matching function

Let's recall that the job finding rate $F_t = \frac{H_t}{U_t}$

The state space representation in this case is the following:

$$F_t = M_t + \rho(V_t - U_t) + \epsilon_t \quad \epsilon_t \sim N(0, R) \quad (30)$$

$$M_t = M_{t-1} + v_t \quad v_t \sim N(0, Q) \quad (31)$$

where also, $\text{Cov}(\epsilon_t, v_t) = 0$ and $(V_t - U_t)$ the labor market tightness.

First Step:

What to do

1 - In the *Main_File.m* set the initial values for the three estimated parameters (Q , R , ρ). (Think of OLS as starting point).

2 - Next, in *Loglikelihood.m* specify the estimated parameters in terms of the notation used in previous section. That is, determine the values of Q , R , C , F , M and A given the input of the minimization routine.

3 - Finally, complete the Kalman filter recursions in *KalmanFilter.m* Then run the program by commenting out the rest of the code and see what happened to match efficiency during the crisis.

References

References

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